you have enormous flexibility (as you'll see in this and later chapters) that allows you to make diagrams and charts that you can't find in other libraries.

Listing 5.1 shows the code to create a histogram layout and associate it with a particular scale. I've also included an example of how you can use interactivity to adjust the original layout and rebind the data to your shapes. This changes the histogram from showing the number of tweets that were favorited to the number of tweets that were retweeted.

Listing 5.1 Histogram code

```
d3.json("tweets.json", function(error, data) { histogram(data.tweets) });

function histogram(tweetsData) {

   var xScale = d3.scale.linear().domain([ 0, 5 ]).range([ 0, 500 ]);
   var yScale = d3.scale.linear().domain([ 0, 10 ]).range([ 400, 0 ]);

   var xAxis = d3.svg.axis().scale(xScale).ticks(5).orient("bottom");

   var histoChart = d3.layout.histogram();

   histoChart.bins([ 0, 1, 2, 3, 4, 5 ]).value(function(d) {
         return d.favorites.length;
     });

   histoData = histoChart(tweetsData);

   d3.select("svg").selectAll("rect").data(histoData).enter()
             .append("rect").attr("x", function(d) {
                   return xScale(d.x);
             }).attr("y", function(d) {
                   return yScale(d.y);
             }).attr("width", xScale(histoData[0].dx) - 2)
             .attr("height", function(d) {
                   return 400 - yScale(d.y);
             }).on("click", retweets);

   d3.select("svg").append("g").attr("class", "x axis")
             .attr("transform", "translate(0,400)").call(xAxis);

   d3.select("g.axis").selectAll("text").attr("dx", 50);

   function retweets() {
        histoChart.value(function(d) {
             return d.retweets.length;
        });

        histoData = histoChart(tweetsData);

        d3.selectAll("rect").data(histoData)
             .transition().duration(500).attr("x", function(d) {
                  return xScale(d.x)
             }).attr("y", function(d) {
                  return yScale(d.y)
             }).attr("height", function(d) {
                  return 400 - yScale(d.y);
             });
   };
};
```

Creates a new layout function

Determines the values the histogram bins for

The value the layout is binning for from the datapoint

Formats the data

Formatted data is used to draw the bars

Centers the axis labels under the bars

Changes the value being measured

Binds and redraws the new data